TRIBHUVAN UNIVERSITY

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HIMALAYA COLLEGE OF ENGINEERING

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Lab Report No:- 06

Title:- IMPLEMENTING Diffie hellman key exchange algorithm, RSA algorithm and Elgamal public key cryptosystem.

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Date Of Submission:- 2082/12/

LAB6: Implementation of Diffie hellman key exchange algorithm, RSA algorithm and Elgamal public key cryptosystem.

Objective:

- To understand & implement the Diffie-Heuman key exchange algorithm.
- To study and implement the RSA public key cryptosystem.
- To implement the Elgamal encryption & decryption.

Theory:

1. Diffie-Heuman key exchange Algorithm

It is a method used to securely exchange cryptographic keys over a public channel.

Steps:

1. select a large prime number p & a primitive root g.
2. user A select private key a, computes:

$$A = g^a \bmod p.$$

3. user B selects Private key b, computes:

$$B = g^b \bmod p.$$

4. Exchange values A & B.

5. compute shared key:

$$K = B^a \bmod p = A^b \bmod p.$$

2. RSA Algorithm

RSA is a widely used public key cryptosystem used for secure data communication.

Steps:

1. choose two large prime numbers p & q.
2. compute:

$$n = p \times q$$
$$\phi(n) = (p-1)(q-1)$$

3. Choose e such that:

$$1 < e < \phi(n),\ \gcd(e, \phi(n)) = 1$$

4. compute private key d:

$$d \cdot e \equiv 1 \pmod{\phi(n)}$$

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Encryption:

$$C = M^e \bmod n$$

Decryption:

$$M = C^d \bmod n$$

3. Elgamal public key cryptosystem

It is an asymmetric encryption algorithm based on the Diffie-Hellman concept.

Steps:

1. Select prime p & primitive root g.
2. Choose private key x.
3. Compute public key:

$$y = g^x \bmod p$$

a. Encryption:

1. Choose random integer k.
2. Compute:

$$C_1 = g^k \bmod p$$

$$C_2 = M \cdot y^k \bmod p$$

Decryption:

$$M = C_2 \cdot (C_1^x)^{-1} \bmod p.$$

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5.1) implement Diffie Hellman key exchange algorithm.

```
#include<stdio.h>
#include <conio.h>
#include<math.h>
int modulo (int a, int b, int mod)
{ int res = 1;
 while (b>0){
   if(b&1)
     res=(res*a) % mod;
     a = (a*a ) % mod;
     b >>= 1;
  }
 return res;
 }
int main(){
  int p,g, xa,ya,xb,yb,ka,kb;
  printf("Enter prime number (p): ");
  scanf("%d", &p);
  printf("Enter primitive root of p:");
  scanf("%d", &g);
  printf("Enter the random number Xa<p:");
  scanf("%d", &xa);
  printf("Enter the random number Xb<p:");
  scanf("%d", &xb);
  ya = modulo(g,xa,p);
  yb=modulo( g, xb,p);
  printf("The private key of user A is %d", ya);
  printf("The private key of user B is %d", yb);
  ka = modulo (yb, xa,p);
  kb = modulo (ya,xb, p);
  printf("The key for user a is %d\n", ka);
  printf("The key for user b is %d", kb);

 }
```

```
for (int i=0; i<k-1; i++){
    x=(x*x)%n;
    Printf("Step %d ;x=x^2 modn= %d\n",i+3,x);
  if (x==n-1){
    Printf("\n Result: %d is Prime\n",n);
    Printf(" Name : Anil Pal Roll no:5");
   return 0;
   }
 }
 Printf("\n Result : %d is not prime\n", n);
 Printf(" Name: Anil Pal Roll no: 5");
 return 0;
}
```

```
C:\Users\Dell\OneDrive\Desk1

Enter prime number (p):29
Enter primitive root of p:19
Enter the random number Xa < p:16
Enter the random number Xb < p:12
The Private key of User A is 16.
The Private key of User B is 20.
The key for user a is 23.
The key for user b is 23.
Process returned 0 (0x0)   execution time : 11.513 s
Press any key to continue.
```

Q.2) C program to implement RSA algorithm.

```
#include<stdio.h>
#include<stdlib.h>
#include<math.h>
int gcd(int a, int b){
  if (b==0) return a;
  return gcd(b, a%b);
}
int modulo(int a, int b, int mod){
  int res=1;
  while(b>0){
  if (b&1) res=(res*a)%mod;
    a=(a*a)%mod;
    b>>=1;
  }
  return res;
  }

int main()
{int p,q,n,phi,e,d,message;
  printf("enter two prime numbers:");
  scanf("%d %d", &p, &q);

  n=p*q;
  phi=(p-1)*(q-1);

for(e=2; e<phi; e++){
    if(gcd(e, phi)==1) break;
  }
for(d=2; d<phi; d++){
    if((e*d)%phi==1) break;
  }
  printf("in public key(n,e)=(%d, %d\n", n, e);
  printf("private key (n,d)=(%d, %d)\n", n, d);

  printf("enter the message (integer<%d):", n);
  scanf("%d", &message);
```

```
int cipher=modulo(message, e,n);
printf("Encrypted message: %d\n", cipher);

int decrypted = modulo (cipher, d,n);
 printf("Decrypted message: %d\n", decrypted);

printf("Name: Anil Pal  Roll no: 5\n");

return 0;
}
```

```
C:\Users\Dell\OneDrive\Desk1 ×  +  ~

Enter two prime numbers: 5 19

Public key (n, e) = (95, 5)
Private key (n, d) = (95, 29)
Enter the message (integer < 95): 28
Encrypted message: 73
Decrypted message: 28
Name: Anil Pal   Roll no: 5

Process returned 0 (0x0)   execution time : 18.149 s
Press any key to continue.
```

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Q.3) Program to implement Elgamal public key cryptosystem.

```
#include<stdio.h>
# include<stdlib.h>
# include<math.h>
int gcd (int a, int b){
   if (b==0){
  return a;
  }
   return gcd (b, a%b);
}
int modulo (int a, int b, int mod){
   int res=1;
while (b>0){
  if (b & 1){
    res=(res * a)% mod;
    }
  a= (a * a)% mod;
  b >>= 1;
  }
return res;
}
int main(){
  int p, g, x, k, message;
   printf("Enter a prime number:");
  scanf("%d", &p);

  printf("Enter a primitive root of %d:", p);
  scanf("%d", &g);

  printf(" enter a private key:");
  scanf("%d", &x);

  printf(" Enter a value for k (1<k< %d)", p-1);
  scanf("%d", &k);
```

```
int y = modulo(g, x, p);
Printf("Enter the message:");
scanf("%d", &message);
int a = modulo(g, k, p);
printf("Encrypted message: (%d, %d)\n", a, b);
int decrypted = (b * modulo(a, p-1-x, p)) % p;
Printf("Decrypted message: %d\n", decrypted);
printf("Name: Anil Pal Roll no: 5");
return 0;
}
```

```
C:\Users\Dell\OneDrive\Desk1
Enter a prime number: 29
Enter a primitive root of 29: 19
Enter a private key: 12
Enter a value for k (1 < k < 28): 18
Enter the message: 9
Encrypted message: (5, 5)
Decrypted message: 9
Name:Anil Pal  Roll no:5
Process returned 0 (0x0)   execution time : 14.770 s
Press any key to continue.
```

DISCUSSION

In this lab, we implemented three important public key cryptographic algorithms. Diffie-Hellman enables two parties to establish a shared secret key over an insecure channel without prior communication. RSA algorithm provides both encryption & decryption using a pair of public & private keys. And, Elgamal cryptosystem is an extension of Diffie-Hellman & provides encryption as well as decryption. It introduce randomness during encryption, making it more secure against certain attacks compared to deterministic algorithm like RSA.

CONCLUSION

In the last of this lab, we were able to implement Diffie Hellman, RSA and Elgamal Algorithm in C programming.

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